SAT Exam Simulator for Moodle (`mod_sat`)

> **Digital SAT® Practice & Multi-Stage Adaptive Testing Engine for Moodle 5.2+**
>
> العربية (النسخة العربية ودليل المعلم) | Technical Plan

Overview

SAT Exam Simulator (`mod_sat`) is a native, production-ready Moodle 5.2+ activity plugin designed for high schools, test preparation centers, and educators. It delivers computer-based digital SAT practice exams featuring **Multi-Stage Adaptive Testing (MST)**, realistic timing and pacing, built-in self-hosted math tools, comprehensive student diagnostic score reports, and class-wide cohort analytics.

Key Highlights

- **Multi-Stage Adaptive Testing (MST)**: Accurately mirrors digital SAT routing: student performance on Module 1 dynamically routes them to either an **Easy** or **Hard** Module 2.
- **Official-Style Scoring (400–1600)**: Scaled 200–800 section scores calculated with authentic psychometric tables and automatic sync to the Moodle Gradebook.
- **Flexible Pacing Modes**:
  - **Continuous Sitting**: Official mock test simulation running all modules back-to-back with a scheduled 10-minute break.
  - **Paced / Multi-Session**: Ideal for weekly homework and modular practice. Students complete Module 1, pause, and unlock their adaptively routed Module 2 later.
- **100% Self-Hosted & Open Source**:
  - Built-in draggable **Graphing & Scientific Calculator** (HTML5 Canvas + JavaScript math engine). Zero commercial API dependencies (no Desmos API key required).
  - Built-in SVG **Math Reference Sheet** containing all standard formulas.
  - Interactive **Option Eliminator** (ABC strike-out) and live **Student-Produced Response (SPR)** validation.
- **Moodle 5.2 Native Architecture**:
  - Full compatibility with Moodle 5.2's Question Bank architecture (`mdl_question_references`).
  - Native Moodle Question Engine integration (`question_usage_by_activity`).
  - Complete Accommodations/Overrides system (1.5x, 2.0x time multipliers).
  - Full Moodle 2 Backup & Restore integration and GDPR Privacy API compliance.
- **Diagnostic Score Reports & Cohort Analytics**:

- Student score reports mapped to the **8 official SAT content domains**.
- Teacher dashboard with score distributions, routing rates, and class-wide domain weakness heatmaps for targeted classroom remediation.

SAT 101: A Beginner's Primer for Teachers

If you or your department are new to the digital SAT, here is how the exam works:

```
[ Section 1: Reading & Writing ]
   ├── Module 1 (Routing Module - Standard mix of questions)
   └── Module 2 (Adaptive Stage)
          ├── Easy Module 2 (For scores below threshold)
          └── Hard Module 2 (For scores at or above threshold)
   [ Optional 10-Minute Break in Full Exam Mode ]
[ Section 2: Math ]
   ├── Module 1 (Routing Module - Standard mix of questions)
   └── Module 2 (Adaptive Stage)
          ├── Easy Module 2 (For scores below threshold)
          └── Hard Module 2 (For scores at or above threshold)
```

1. Multi-Stage Adaptive Testing (MST)

Unlike linear tests where every student answers identical questions, the digital SAT adapts between modules:

- Every student starts with **Module 1** (a balanced mix of easy, medium, and hard questions).
- The student's score in Module 1 determines their route:
  - If they achieve $\geq 60\%$ (configurable), they are routed to **Module 2 Hard**. Routing to Hard unlocks the highest score bands (up to 800 per section).
  - If they achieve $< 60\%$, they are routed to **Module 2 Easy**. Scores are capped to lower bands (typically 200–600).

2. Scoring Structure

- **Reading and Writing**: Scaled from **200 to 800**.
- **Math**: Scaled from **200 to 800**.
- **Total Composite Score**: Sum of both sections, ranging from **400 to 1600**.

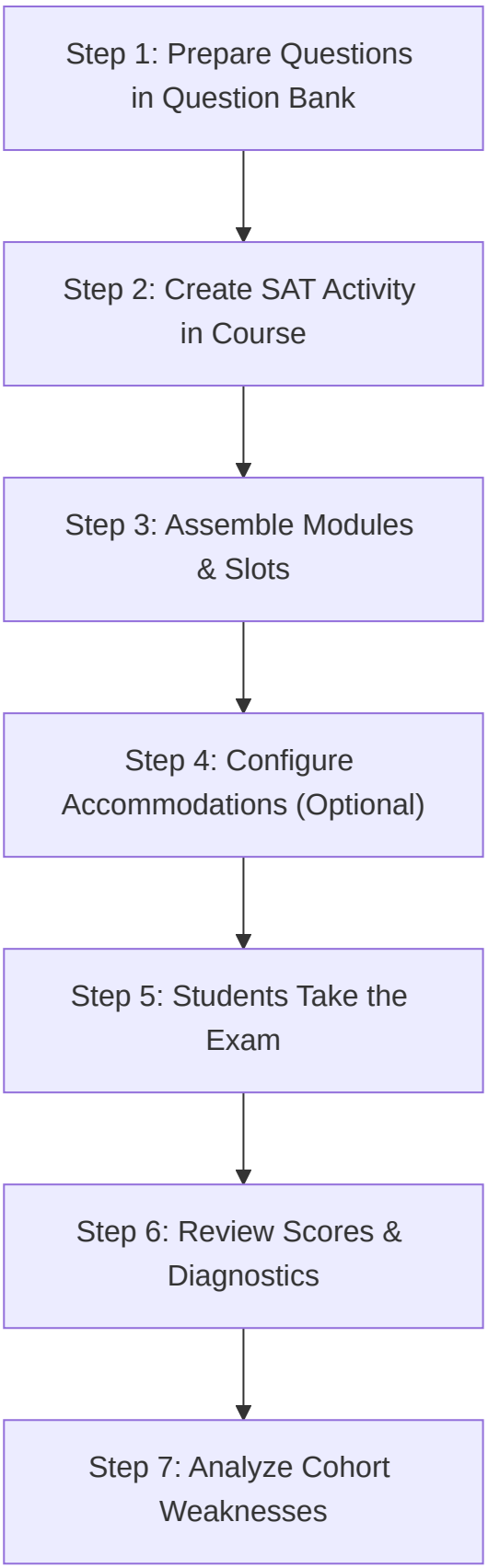
3. Question Formats

- **Multiple Choice (MCQ)**: 4 options (A, B, C, D) with a single correct choice.

- **Student-Produced Response (SPR / Grid-ins)**: Open-ended numerical or fractional answers in the Math section (e.g. `2/3`, `0.667`, `-4/5`).

Step-by-Step Teacher Wizard Guide

Follow these 7 steps to configure and run your first SAT practice exam:



Step 1: Prepare Questions in Moodle's Question Bank

Before creating your exam activity, create your questions in Moodle's **Question Bank** (`Course > Question bank`):

A. Multiple-Choice Questions (MCQ)

1. Add a standard **Multiple Choice** question.
2. Select **One answer only**.
3. Provide 4 answer choices (A, B, C, D) and assign 100% grade to the correct choice.

B. Math Student-Produced Responses (SPR / Grid-in)

1. Add a **Short answer** or **Numerical** question.
2. Enter the canonical answer (e.g., `2/3` or `0.75`).
3. The plugin automatically accepts equivalent forms:

   - Fractions and decimals: `2/3`, `.6666`, `.6667`, `0.666`, `0.667` are all recognized as correct.

   - Negative numbers (up to 6 characters, e.g. `-10.5` or `-7/12`).

   - Positive numbers (up to 5 characters, e.g. `3.141` or `15/16`).

C. Formatting Split-Screen Reading Passages

For Reading & Writing questions, place a horizontal rule (`<hr>` or the **Horizontal line** button in the Atto/TinyMCE editor) between your passage and the question stem:

```
<p>In 1928, Alexander Fleming observed a mold that killed bacteria...</p>
<hr>
<p>Which choice best describes the main idea of the passage?</p>
```

The exam simulator automatically splits this into a **two-column layout**: the passage appears on the left, and the question with choices appears on the right.

D. Tagging Questions with SAT Domains

To unlock student diagnostic reports and cohort analytics, add a tag to each question corresponding to the official SAT domain:

- Reading & Writing:
  - `sat_domain:craft_structure` (Words in Context, Text Structure and Purpose)
  - `sat_domain:information_ideas` (Central Ideas, Command of Evidence)
  - `sat_domain:standard_english` (Boundaries, Form, Structure, and Sense)

    - `sat_domain:expression_ideas` (Rhetorical Synthesis, Transitions)
- Math:
    - `sat_domain:algebra` (Linear equations, Linear inequalities, Systems)
    - `sat_domain:advanced_math` (Nonlinear functions, Quadratics, Polynomials)
    - `sat_domain:problem_solving_data` (Ratios, Rates, Percentages, Probability)
    - `sat_domain:geometry_trig` (Area, Volume, Right triangles, Circle theorems)

Step 2: Create a New SAT Activity in Your Course

1. Turn on **Edit mode** in your Moodle course.
2. Click **Add an activity or resource** and select **SAT Exam Simulator**.
3. Configure the settings:

Setting	Recommendation	Description
Exam Name	*Official Practice Exam 1*	Activity title shown on the course page.
Exam Scope	**Full SAT Exam**	Choose *Full SAT* (RW + Math), *Math Only*, *RW Only*, or *Single Module Practice*.
Session Pacing	**Continuous Sitting**	Choose *Continuous* for realistic timed mock exams, or *Paced / Multi-Session* for modular homework assignments.
Routing Mode	**Adaptive (MST)**	Keeps official adaptive routing enabled.
Routing Threshold	**60%** (Default)	Percentage score on Module 1 required to route student to Hard Module 2.
Timing	32 min/module (RW) 35 min/module (Math)	Standard official timing per module.
Attempts Allowed	1 (or more)	Maximum attempts a student can make.
Grading Method	Highest grade	Aggregation method across multiple attempts.

4. Click **Save and display**.

Step 3: Assemble Modules and Assign Questions

On the activity page, click **Manage Questions** (`edit.php`):

1. You will see tabs for each module depending on the chosen exam scope:

   - **Reading & Writing: Module 1 (Routing)**

   - **Reading & Writing: Module 2 (Easy)**

   - **Reading & Writing: Module 2 (Hard)**

   - **Math: Module 1 (Routing)**

   - **Math: Module 2 (Easy)**

   - **Math: Module 2 (Hard)**

2. Select a module tab.

3. In the **Add Question Slot** form at the bottom:

   - Select a question from your Question Bank.

   - Set the point value (default: `1.00`).

   - Click **Add slot**.

4. Use the arrow icons (`↑` / `↓`) to rearrange questions, or `×` to remove slots.

5. Standard question counts for an official SAT test:

   - Reading & Writing: **27 questions per module** (54 total across 2 modules).

   - Math: **22 questions per module** (44 total across 2 modules).

Step 4: Configure Accommodations & Overrides (Optional)

If you have students with IEP, 504, or testing accommodations (such as extended time):

1. On the activity page, click **Overrides and Accommodations** (`overrides.php`).

2. Click **Add user accommodation override**.

3. Select the student and configure their accommodation:

   - **Time multiplier**: - `1.0x` : Standard time. - `1.5x` : 50% extended time (Time and a half). - `2.0x` : 100% extended time (Double time).

   - **Attempts allowed**: Grant additional attempts if required.

4. Click **Save override**.

Step 5: How Students Experience the Exam Simulator

When students start an attempt, they enter a distraction-free, full-screen exam runner:

- **Top Navigation Bar**:
  - **Countdown Timer**: Server-authoritative timer. When remaining time drops below 5 minutes, the timer turns red with an alert. Students can click *Hide* if they prefer to reduce anxiety.
  - **Reference Sheet**: Opens a modal with all official Math geometric and algebraic formulas.
  - **Calculator**: Opens a draggable scientific and graphing calculator.
- **Split-Screen Layout**:
  - Passages appear on the left column with clean typography and scrolling.
  - Questions, answer options, and navigation buttons appear on the right column.
- **Option Eliminator**:
  - Students can toggle the Option Eliminator (ABC) button to strike through options they have ruled out.
- **Student-Produced Response (SPR) Live Validator**:
  - Numerical inputs only permit valid mathematical characters (`0-9`, `.`, `/`, `-`).
  - Real-time feedback alerts students if a fraction or decimal exceeds character length.
- **Question Matrix Drawer**:
  - Students can open the drawer to view all questions at a glance, jump to any question, and flag questions for review before submitting the module.

Step 6: Grading, Review & Student Diagnostic Reports

1. **Automatic Scoring & Gradebook Sync**:

   - As soon as a student completes their exam, raw scores are converted into official scaled scores (200–800 section, 400–1600 composite).

   - The final score is immediately posted to the Moodle Gradebook (`grade_update`).

2. **Student Diagnostic Score Report (`diagnostic.php`)**:

   - Students and teachers can view a comprehensive score card: - Reading & Writing scaled score (out of 800). - Math scaled score (out of 800). - Total composite score (out of 1600). - **8 Domain Mastery Bars**: Displays accuracy percentages and color-coded mastery badges

(**Mastered**, **Proficient**, or **Needs Practice**) across all SAT domains. - One-click **Print Score Report** button for parent-teacher conferences.

3. **Question-by-Question Review (`review.php`)**:

   - Review every question with the student's selected answer, the correct answer, and full teacher explanations.

Step 7: Teacher Cohort Analytics Dashboard

Teachers can access deep classroom insights by clicking **Cohort Analytics** (`analytics.php`):

1. **Cohort KPI Cards**:

   - Total finished submissions.

   - Class average composite score, highest score, and lowest score.

   - Section averages for Reading & Writing and Math.

2. **Score Distribution Chart**:

   - Visual breakdown across SAT score bands: - `1400–1600`: Highly Competitive / Ivy League benchmark. - `1200–1390`: Competitive / Honors benchmark. - `1000–1190`: National Average / Proficient. - `800–990`: Developing. - `400–790`: Foundational remediation required.

3. **Adaptive Routing Performance**:

   - Reading & Writing: % of students routed to Hard Module 2 vs Easy Module 2.

   - Math: % of students routed to Hard Module 2 vs Easy Module 2.

4. **Domain Weakness Heatmap**:

   - Color-coded classroom accuracy across all 8 SAT domains.

   - Instantly highlights domains requiring immediate lesson review and targeted practice.

Technical Specifications

- **Moodle Compatibility**: Moodle 5.2 and higher (Moodle 5.2, 5.3, etc.).
- **PHP Requirements**: PHP 8.3 or PHP 8.4+.
- **Database Support**: MySQL 8.4+, MariaDB 10.11+, PostgreSQL 16+.

- **Zero Third-Party API Reliance**: Graphing calculator and formula reference sheets run 100% in-browser on client JavaScript and HTML5 Canvas.
- **Privacy & GDPR**: Implements `core_privacy\local\metadata\provider` and `core_privacy\local\request\plugin\provider` for full data export and erasure.
- **Backup & Restore**: Native Moodle 2 XML backup and restore handlers for seamless course imports/exports.

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